Algebra Mastery Pack — Practice Test C

Grade 11 algebra, functions & data analysis · 24 questions · suggested time 45 minutes · no calculator needed

Name: _____ Date: _____
Score: _____ / 24

Instructions. Write full working in the space under each question; answers alone do not show how you got there. Circle one letter for a multiple-choice question and write a single value on the answer line for a short-answer question. Answer every question — a blank scores zero. The answer key with a worked solution for every question is a separate PDF; mark your own paper only after you finish.

1. [Solving Linear Equations (for x)]

Solve $7x = 42$. Enter the value of x.

Answer: _____

2. [Literal Equations & Solving for y]

Solve $P = 2l + 2w$ for w.

A) $w = (P - 2l) \div 2$
B) $w = P - 2l$
C) $w = (P + 2l) \div 2$
D) $w = 2P - 2l$

3. [Linear Inequalities]

Solve $3x - 6 \geq 9$.

A) $x \geq 5$
B) $x \leq 5$
C) $x \geq 15$
D) $x \leq 15$

4. [Systems of Linear Equations]

Solve by elimination: $2x + y = 8$ and $x - y = 1$.

A) (3, 2)
B) (2, 3)
C) (3, 3)
D) (1, 6)

5. [Functions: Notation, Domain & Range]

If $h(x) = 5x - 4$, what is $h(2)$? Enter a number.

Answer: _____

6. [Linear Functions: Slope, Intercepts & Equations]

Find the slope through the points (1, 2) and (3, 8). Enter a number.

Answer: _____

7. [Quadratic Functions & Equations]

The graph of $y = -x^2 + 3$ opens ___.

A) downward
B) upward
C) left
D) right

8. [Polynomials: Operations & Factoring]

Factor out the greatest common factor from $6x + 12$. Enter the GCF.

Answer: _____

9. [Radical & Rational Expressions]

To solve a rational equation, multiply both sides by the ___.

A) least common denominator (LCD)
B) greatest common factor
C) numerator
D) variable

10. [Exponential Functions & Growth/Decay]

For $y = 100 \cdot (1/2)^x$, find y when x = 2. Enter a number.

Answer: _____

11. [Data: Measures of Center & Spread]

Which measure of center is most affected by an outlier (one very large or small value)?

A) Mean
B) Median
C) Mode
D) Range

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12. [Scatter Plots, Line of Best Fit & Correlation]

A correlation coefficient r near -1 indicates ___.

A) strong negative correlation
B) strong positive correlation
C) no correlation
D) an error

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13. [Absolute Value Equations & Inequalities]

Solve $|2x + 1| = 9$. Enter the larger of the two solutions.

Answer: _____

14. [Arithmetic & Geometric Sequences]

What is the common ratio of the geometric sequence 2, 6, 18, 54, ...?

A) 2
B) 3
C) 4
D) 12

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15. [Inverse Functions]

For $f(x) = x^2$ with domain $x \geq 0$, what is $f^{-1}(x)$?

A) $\sqrt{x}$
B) x^2
C) $1/x^2$
D) $-\sqrt{x}$

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16. [Function Transformations & Vertex Form]

What does $y = -f(x)$ do to the graph of $y = f(x)$?

A) reflects it across the x-axis
B) reflects it across the y-axis
C) shifts it 1 unit down
D) rotates it 90°

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17. [Systems of Linear Inequalities & Linear Programming]

A feasible region has vertices (0, 0), (5, 0), (0, 4) and (2, 3). Where is $P = 3x + 2y$ maximised?

A) (0, 0)
B) (5, 0)
C) (0, 4)
D) (2, 3)

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18. [Probability from Two-Way Tables]

What is P(walks and is late)?

A) 0.15
B) 0.25
C) 0.60
D) 0.30

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19. [Quadratic Functions & Equations]

For $y = x^2 - 4x + 3$, the vertex's x-coordinate is $x = -b \div (2a)$. Enter that value.

Answer: _____

20. [Absolute Value Equations & Inequalities]

Solve $|x - 4| = -2$.

A) $x = 6$
B) $x = 2$
C) $x = 6$ or $x = 2$
D) No solution

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21. [Arithmetic & Geometric Sequences]

A geometric sequence starts 5, 10, 20, … What is its 6th term?

Answer: _____

22. [Inverse Functions]

Which function has an inverse that is also a function?

A) $f(x) = x^2$ for all real x
B) $f(x) = |x|$ for all real x
C) $f(x) = 2x + 5$
D) $f(x) = 4$ (a constant)

23. [Function Transformations & Vertex Form]

$y = f(2x)$ is a…

A) horizontal compression by a factor of 1/2
B) horizontal stretch by a factor of 2
C) vertical compression by 1/2
D) vertical stretch by 2

24. [Systems of Linear Inequalities & Linear Programming]

A region is bounded by $x \geq 0$, $y \geq 0$ and $x + y \leq 10$. What is the largest value of $P = 2x + 3y$ on that region?

Answer: _____